

Corollary 2.7 *A power series is infinitely complex differentiable in its disc of convergence, and the higher derivatives are also power series obtained by termwise differentiation.*

We have so far dealt only with power series centered at the origin. More generally, a power series centered at $z_0 \in \mathbb{C}$ is an expression of the form

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

The disc of convergence of f is now centered at z_0 and its radius is still given by Hadamard's formula. In fact, if

$$g(z) = \sum_{n=0}^{\infty} a_n z^n,$$

then f is simply obtained by translating g , namely $f(z) = g(w)$ where $w = z - z_0$. As a consequence everything about g also holds for f after we make the appropriate translation. In particular, by the chain rule,

$$f'(z) = g'(w) = \sum_{n=0}^{\infty} n a_n (z - z_0)^{n-1}.$$

A function f defined on an open set Ω is said to be **analytic** (or have a **power series expansion**) at a point $z_0 \in \Omega$ if there exists a power series $\sum a_n(z - z_0)^n$ centered at z_0 , with positive radius of convergence, such that

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n \quad \text{for all } z \text{ in a neighborhood of } z_0.$$

If f has a power series expansion at every point in Ω , we say that f is **analytic on Ω** .

By Theorem 2.6, an analytic function on Ω is also holomorphic there. A deep theorem which we prove in the next chapter says that the converse is true: every holomorphic function is analytic. For that reason, we use the terms holomorphic and analytic interchangeably.

3 Integration along curves

In the definition of a curve, we distinguish between the one-dimensional geometric object in the plane (endowed with an orientation), and its

parametrization, which is a mapping from a closed interval to $\mathbb{C}$, that is not uniquely determined.

A **parametrized curve** is a function $z(t)$ which maps a closed interval $[a, b] \subset \mathbb{R}$ to the complex plane. We shall impose regularity conditions on the parametrization which are always verified in the situations that occur in this book. We say that the parametrized curve is **smooth** if $z'(t)$ exists and is continuous on $[a, b]$, and $z'(t) \neq 0$ for $t \in [a, b]$. At the points $t = a$ and $t = b$, the quantities $z'(a)$ and $z'(b)$ are interpreted as the one-sided limits

$$z'(a) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{z(a+h) - z(a)}{h} \quad \text{and} \quad z'(b) = \lim_{\substack{h \rightarrow 0 \\ h < 0}} \frac{z(b+h) - z(b)}{h}.$$

In general, these quantities are called the right-hand derivative of $z(t)$ at a , and the left-hand derivative of $z(t)$ at b , respectively.

Similarly we say that the parametrized curve is **piecewise-smooth** if z is continuous on $[a, b]$ and if there exist points

$$a = a_0 < a_1 < \cdots < a_n = b,$$

where $z(t)$ is smooth in the intervals $[a_k, a_{k+1}]$. In particular, the right-hand derivative at a_k may differ from the left-hand derivative at a_k for $k = 1, \dots, n-1$.

Two parametrizations,

$$z : [a, b] \rightarrow \mathbb{C} \quad \text{and} \quad \tilde{z} : [c, d] \rightarrow \mathbb{C},$$

are **equivalent** if there exists a continuously differentiable bijection $s \mapsto t(s)$ from $[c, d]$ to $[a, b]$ so that $t'(s) > 0$ and

$$\tilde{z}(s) = z(t(s)).$$

The condition $t'(s) > 0$ says precisely that the orientation is preserved: as s travels from c to d , then $t(s)$ travels from a to b . The family of all parametrizations that are equivalent to $z(t)$ determines a **smooth curve** $\gamma \subset \mathbb{C}$, namely the image of $[a, b]$ under z with the orientation given by z as t travels from a to b . We can define a curve γ^- obtained from the curve γ by reversing the orientation (so that γ and γ^- consist of the same points in the plane). As a particular parametrization for γ^- we can take $z^- : [a, b] \rightarrow \mathbb{R}^2$ defined by

$$z^-(t) = z(b + a - t).$$

It is also clear how to define a **piecewise-smooth curve**. The points $z(a)$ and $z(b)$ are called the **end-points** of the curve and are independent on the parametrization. Since γ carries an orientation, it is natural to say that γ begins at $z(a)$ and ends at $z(b)$.

A smooth or piecewise-smooth curve is **closed** if $z(a) = z(b)$ for any of its parametrizations. Finally, a smooth or piecewise-smooth curve is **simple** if it is not self-intersecting, that is, $z(t) \neq z(s)$ unless $s = t$. Of course, if the curve is closed to begin with, then we say that it is simple whenever $z(t) \neq z(s)$ unless $s = t$, or $s = a$ and $t = b$.

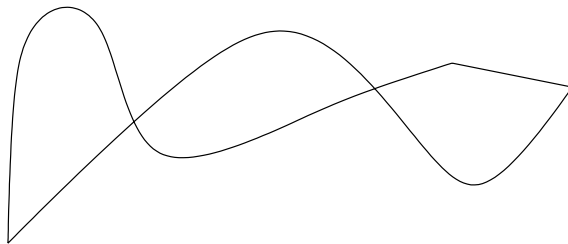


Figure 3. A closed piecewise-smooth curve

For brevity, we shall call any piecewise-smooth curve a **curve**, since these will be the objects we shall be primarily concerned with.

A basic example consists of a circle. Consider the circle $C_r(z_0)$ centered at z_0 and of radius r , which by definition is the set

$$C_r(z_0) = \{z \in \mathbb{C} : |z - z_0| = r\}.$$

The **positive orientation** (counterclockwise) is the one that is given by the standard parametrization

$$z(t) = z_0 + re^{it}, \quad \text{where } t \in [0, 2\pi],$$

while the **negative orientation** (clockwise) is given by

$$z(t) = z_0 + re^{-it}, \quad \text{where } t \in [0, 2\pi].$$

In the following chapters, we shall denote by C a general *positively* oriented circle.

An important tool in the study of holomorphic functions is integration of functions along curves. Loosely speaking, a key theorem in complex

analysis says that if a function is holomorphic in the interior of a closed curve γ , then

$$\int_{\gamma} f(z) dz = 0,$$

and we shall turn our attention to a version of this theorem (called Cauchy's theorem) in the next chapter. Here we content ourselves with the necessary definitions and properties of the integral.

Given a smooth curve γ in $\mathbb{C}$ parametrized by $z : [a, b] \rightarrow \mathbb{C}$, and f a continuous function on γ , we define the **integral of f along γ** by

$$\int_{\gamma} f(z) dz = \int_a^b f(z(t))z'(t) dt.$$

In order for this definition to be meaningful, we must show that the right-hand integral is independent of the parametrization chosen for γ . Say that $\tilde{z}$ is an equivalent parametrization as above. Then the change of variables formula and the chain rule imply that

$$\int_a^b f(z(t))z'(t) dt = \int_c^d f(z(t(s)))z'(t(s))t'(s) ds = \int_c^d f(\tilde{z}(s))\tilde{z}'(s) ds.$$

This proves that the integral of f over γ is well defined.

If γ is piecewise smooth, then the integral of f over γ is simply the sum of the integrals of f over the smooth parts of γ , so if $z(t)$ is a piecewise-smooth parametrization as before, then

$$\int_{\gamma} f(z) dz = \sum_{k=0}^{n-1} \int_{a_k}^{a_{k+1}} f(z(t))z'(t) dt.$$

By definition, the **length** of the smooth curve γ is

$$\text{length}(\gamma) = \int_a^b |z'(t)| dt.$$

Arguing as we just did, it is clear that this definition is also independent of the parametrization. Also, if γ is only piecewise-smooth, then its length is the sum of the lengths of its smooth parts.

Proposition 3.1 *Integration of continuous functions over curves satisfies the following properties:*

(i) *It is linear, that is, if $\alpha, \beta \in \mathbb{C}$, then*

$$\int_{\gamma} (\alpha f(z) + \beta g(z)) dz = \alpha \int_{\gamma} f(z) dz + \beta \int_{\gamma} g(z) dz.$$

(ii) *If γ^- is γ with the reverse orientation, then*

$$\int_{\gamma} f(z) dz = - \int_{\gamma^-} f(z) dz.$$

(iii) *One has the inequality*

$$\left| \int_{\gamma} f(z) dz \right| \leq \sup_{z \in \gamma} |f(z)| \cdot \text{length}(\gamma).$$

Proof. The first property follows from the definition and the linearity of the Riemann integral. The second property is left as an exercise. For the third, note that

$$\left| \int_{\gamma} f(z) dz \right| \leq \sup_{t \in [a, b]} |f(z(t))| \int_a^b |z'(t)| dt \leq \sup_{z \in \gamma} |f(z)| \cdot \text{length}(\gamma)$$

as was to be shown.

As we have said, Cauchy's theorem states that for appropriate closed curves γ in an open set Ω on which f is holomorphic, then

$$\int_{\gamma} f(z) dz = 0.$$

The existence of primitives gives a first manifestation of this phenomenon. Suppose f is a function on the open set Ω . A **primitive** for f on Ω is a function F that is holomorphic on Ω and such that $F'(z) = f(z)$ for all $z \in \Omega$.

Theorem 3.2 *If a continuous function f has a primitive F in Ω , and γ is a curve in Ω that begins at w_1 and ends at w_2 , then*

$$\int_{\gamma} f(z) dz = F(w_2) - F(w_1).$$

Proof. If γ is smooth, the proof is a simple application of the chain rule and the fundamental theorem of calculus. Indeed, if $z(t) : [a, b] \rightarrow \mathbb{C}$ is a parametrization for γ , then $z(a) = w_1$ and $z(b) = w_2$, and we have

$$\begin{aligned} \int_{\gamma} f(z) dz &= \int_a^b f(z(t)) z'(t) dt \\ &= \int_a^b F'(z(t)) z'(t) dt \\ &= \int_a^b \frac{d}{dt} F(z(t)) dt \\ &= F(z(b)) - F(z(a)). \end{aligned}$$

If γ is only piecewise-smooth, then arguing as we just did, we obtain a telescopic sum, and we have

$$\begin{aligned} \int_{\gamma} f(z) dz &= \sum_{k=0}^{n-1} F(z(a_{k+1})) - F(z(a_k)) \\ &= F(z(a_n)) - F(z(a_0)) \\ &= F(z(b)) - F(z(a)). \end{aligned}$$

Corollary 3.3 *If γ is a closed curve in an open set Ω , and f is continuous and has a primitive in Ω , then*

$$\int_{\gamma} f(z) dz = 0.$$

This is immediate since the end-points of a closed curve coincide.

For example, the function $f(z) = 1/z$ does not have a primitive in the open set $\mathbb{C} - \{0\}$, since if C is the unit circle parametrized by $z(t) = e^{it}$, $0 \leq t \leq 2\pi$, we have

$$\int_C f(z) dz = \int_0^{2\pi} \frac{ie^{it}}{e^{it}} dt = 2\pi i \neq 0.$$

In subsequent chapters, we shall see that this innocent calculation, which provides an example of a function f and closed curve γ for which $\int_{\gamma} f(z) dz \neq 0$, lies at the heart of the theory.

Corollary 3.4 *If f is holomorphic in a region Ω and $f' = 0$, then f is constant.*

Proof. Fix a point $w_0 \in \Omega$. It suffices to show that $f(w) = f(w_0)$ for all $w \in \Omega$.

Since Ω is connected, for any $w \in \Omega$, there exists a curve γ which joins w_0 to w . Since f is clearly a primitive for f' , we have

$$\int_{\gamma} f'(z) dz = f(w) - f(w_0).$$

By assumption, $f' = 0$ so the integral on the left is 0, and we conclude that $f(w) = f(w_0)$ as desired.

Remark on notation. When convenient, we follow the practice of using the notation $f(z) = O(g(z))$ to mean that there is a constant $C > 0$ such that $|f(z)| \leq C|g(z)|$ for z in a neighborhood of the point in question. In addition, we say $f(z) = o(g(z))$ when $|f(z)/g(z)| \rightarrow 0$. We also write $f(z) \sim g(z)$ to mean that $f(z)/g(z) \rightarrow 1$.

4 Exercises

1. Describe geometrically the sets of points z in the complex plane defined by the following relations:

(a) $|z - z_1| = |z - z_2|$ where $z_1, z_2 \in \mathbb{C}$.

(b) $1/z = \bar{z}$.

(c) $\operatorname{Re}(z) = 3$.

(d) $\operatorname{Re}(z) > c$, (resp., $\geq c$) where $c \in \mathbb{R}$.

(e) $\operatorname{Re}(az + b) > 0$ where $a, b \in \mathbb{C}$.

(f) $|z| = \operatorname{Re}(z) + 1$.

(g) $\operatorname{Im}(z) = c$ with $c \in \mathbb{R}$.

2. Let $\langle \cdot, \cdot \rangle$ denote the usual inner product in $\mathbb{R}^2$. In other words, if $Z = (x_1, y_1)$ and $W = (x_2, y_2)$, then

$$\langle Z, W \rangle = x_1x_2 + y_1y_2.$$

Similarly, we may define a Hermitian inner product $(\cdot, \cdot)$ in $\mathbb{C}$ by

$$(z, w) = z\bar{w}.$$